

SHRIRAM CHANDRAN

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EDUCATION

ETH Zürich

Doctor of Sciences Computer Science

since Nov. 2025
Zürich, Switzerland

- Doctoral student at the Scalable Parallel Computing Laboratory.

ETH Zürich

Master of Science in Computer Science

Sep. 2022 – Sep. 2025
Zürich, Switzerland

- **Focus:** Secure & Reliable Systems and Theoretical Computer Science.
- **Thesis:** HO-GDB: Grade 6/6

Indian Institute of Technology Madras

B. Tech. Computer Science and Engineering

Jul. 2018 – Jun. 2022
Chennai, India

- **GPA:** 9.04/10
- **Thesis:** Program Analysis - StarPlat DSL: Grade 9/10

EXPERIENCE

Master Thesis with SPCL

HO-GDB: A Unified Framework for Integrating Higher-Order Structures into Graph Databases

Oct. 2024 - Apr. 2025

ETH Zürich, Switzerland

- Designed algorithms to project higher-order graphs as plain graphs, performance models and proofs of correctness.
- Reasoned about analytical and transactional integrity for higher-order OLTP and OLAP queries.
- Implemented our analysis in HO-GDB, a higher-order graph database with CRUD capabilities.
- Benchmarked HO-GDB across higher-order graph workloads on different datasets and demonstrated scalability.

Practical Work with Systems Group

GPU-Acceleration of the Maximus Database Engine

Oct. 2023 – July. 2024

ETH Zürich, Switzerland

- Implemented GPU support to Maximus, a distributed, scalable and heterogeneous database engine.
- Designed Arrow-compatible data structures to store databases on GPUs for GPU processing on various architectures.
- Integrated GPU-accelerated operators with the existing Maximus interfaces.
- Benchmarked Maximus-GPU on the complete TPC-H query set and demonstrated massive performance gains.

Bachelor Project with Prof. Rupesh Nasre

Program Analysis - StarPlat DSL

Jan. 2022 – Jun. 2022

IIT Madras, India

- Implemented compiler level optimizations for StarPlat, a graph processing DSL.
- Analyzed and implemented multiple optimizations for the DSL compiler using AST-level analyses.
- Conducted successful testing of the implemented optimizations on multiple large graphs across different algorithms.
- Produced positive performance gains for generated code.

Undergraduate Research with Prof. Raghavendra Rao

Stability Analysis

Aug. 2021 – Nov. 2021

IIT Madras, India

- Conducted a comprehensive literature review of previous research on stability of instances for various popular algorithms, in particular Bilu-Linial stability amongst other definitions.
- Analyzed stable instances of Christofides' algorithm under various stability conditions.
- Identified improved approximation bounds for stable instances of metric TSP.
- Developed new approximation algorithms for the vertex cover problem for almost-stable instances in random graphs.

Software Engineering Internship

Rubrik

May 2021 – Jul. 2021

Bangalore, India

- Developed a statistics system with the Blobstore team for analyzing backup metadata job performance.
- Designed a functional model for storing job statistics within their respective databases.
- Enabled access to histograms and statistical data to aid in debugging and data analytics.
- Validated the model on live clusters, demonstrating seamless integration with the main codebase.

Software Engineering Internship

Godot Media

Nov. 2020 – Jan. 2021

Bangalore, India

- Designed and constructed an automatic plagiarism checker to search text against documents on the internet.
- Implemented heuristics and parallelization to increase accuracy and reduce algorithm latency.
- Developed a graphical UI to accept data from a user and display results with highlighted matching text.
- Achieved highly accurate plagiarism detection on articles of up to 2000 words in just a few seconds.

SKILLS

Languages: C/C++, Python, Scala, Haskell, OCaml, Java, C#

Systems / Parallel Computing: CUDA, HIP/ROCm, MPI, PTX, x86-64, ARM64, RISC-V

Frameworks / Tools: Git, Linux, $\LaTeX$, Coq, Django, HTML/CSS, Javascript

Databases / Query Languages: SQL, Cypher

PATENTS AND PUBLICATIONS

- [1] Qimen Xu^{*†}, Yu Zhang^{*}, Dixin Ni, Lei Gao, Guangnan Feng, Qinrui Zheng, Jianting Liu, Haitian Lu, Zhaopeng Jia, Wei Xue, **Shriram Chandran**, Torsten Hoefer[†], Haohuan Fu[†], and Yutong Lu[†]. [Extreme-Scale Linear-Scaling Kohn-Sham DFT at 100 Million Atoms: Bridging Quantum Simulations and Experiments](#). 2026. arXiv: 2609.13115 [cs.CE]. URL: <https://arxiv.org/abs/2609.13115>.
- [2] Maciej Besta^{*†}, Robert Gerstenberger, Konrad Patryński, Thösam Norlha-Tsang, Siran Seifeddini, **Shriram Chandran**, Michal Podstawski, Tao Zhang, Lorenzo Paele, Mischa Orlow, Ales Kubicek, Karol Augustyniak, Hubert Niewiadomski, Marcin Copik, and Torsten Hoefer. [Rigorous Moral Reasoning in Large Language Models](#). Feb. 2026. doi: 10.48550/arXiv.2602.. arXiv: 2602.. URL: <http://arxiv.org/abs/2602..>
- [3] Yakup Budanaz^{*†}, **Shriram Chandran**, Alexandru Calotoiu, and Torsten Hoefer. "The Data Must Flow (To Vector Processors): Searching Program Variants to Improve Compiler Auto-Vectorization Capabilities". In: ().
- [4] Ashwina Kumar^{*†}, Naveen LS, **Shriram Chandran**, Hemesh DJ, Gudla Sanjay, Kovvuri Sraavan Kumar Reddy, Nibedita Behera, and Rupesh Nasre. [Code Generation and Optimization for Different Targets for a Graph DSL](#). Oct. 2025. URL: <http://dx.doi.org/10.22541/au.175987810.03504550/v1>.
- [5] Maciej Besta^{*†}, **Shriram Chandran**, Robert Gerstenberger, Mathis Lindner, Marcin Chrapek, Sebastian Hermann Martschat, Taraneh Ghandi, Patrick Iff, Hubert Niewiadomski, Piotr Nyczyk, Jürgen Müller, and Torsten Hoefer. [Psychologically Enhanced AI Agents](#). 2025. arXiv: 2509.04343 [cs.AI]. URL: <https://arxiv.org/abs/2509.04343>.
- [6] Maciej Besta^{*†}, **Shriram Chandran**^{*}, Jakub Cudak, Patrick Iff, Marcin Copik, Robert Gerstenberger, Tomasz Szydło, Jürgen Müller, and Torsten Hoefer. [Higher-Order Graph Databases](#). 2025. arXiv: 2506.19661 [cs.DB]. URL: <https://arxiv.org/abs/2506.19661>.
- [7] Marko Kabić^{*†}, **Shriram Chandran**, and Gustavo Alonso. [Maximus: A Modular Accelerated Query Engine for Data Analytics on Heterogeneous Systems](#). New York, NY, USA, June 2025. doi: 10.1145/3725324. URL: <https://doi.org/10.1145/3725324>.
- [8] Rupesh Nasre, Ashwina Kumar, Ebenezer Rajadurai, Nibedita Behera, Rajesh Pandian Muniasamy, Shan Mohammed, Sai Nitish, Jash Khatri, Naveen S, **Shriram Chandran**, M. Venkatakrishna, and Rushabh Lalwani. [System and Method for Automatic Parallel Code Generation for Graph Algorithms for Multiple Target Architectures](#). Indian Patent 432922. Granted. Nov. 2022.

SCHOLASTIC ACHIEVEMENTS

JEE: All India Rank **74** in JEE (Advanced) 2018, All India Rank **221** in JEE (Main) 2018

Olympiads: Placed in the top 36 students in India in the 2018 Indian National **Physics**, **Chemistry**, and **Astronomy** Olympiads

KVPY: Two-time Indian Government KVPY fellow (All India Rank **72** in 2016, **68** in 2017)

TEACHING

Student Mentoring

ETH Zürich

- **Dominic Rinderer** *BSc Thesis, Spring 2026*
Research on automated CUTLASS kernel selection.
- **Nicholas Sommer** *MSc Thesis, Spring 2026*
Research on automatic generation of cuTile kernels.
- **Dalia Shaaban** *MSc Thesis, Autumn 2026*
Research on sparse workspaces for GPU sparse computations.

Teaching Assistant

ETH Zürich

since Feb. 2026
Zürich, Switzerland

- 252-0029-00L Parallele Programmierung (Spring Semester 2026)
- 252-0836-00L Informatik II (Autumn Semester 2026)

Coordinator, Programming Club

CFI IIT Madras

Mar. 2019 – Jun. 2020
Chennai, India

- Organized educational sessions on data structures and algorithms, fostering a culture of programming excellence in the institute.
- Organized regular collegiate programming contests to enhance students' programming abilities.

Content Developer

Melvano

Dec. 2018 – Mar. 2019
Chennai, India

- Mentored JEE aspirants by creating Q&A content and addressing doubts through Melvano, an indigenous startup building an educational social platform.
- Helped with the app's initial publicity with UI and design suggestions that enhanced its user appeal.